

Depicting Uncertainty in 2.5D Treemaps | VINCI 2020

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Truth in Visualization

*"[…] what is **not** surrounded by uncertainty cannot be the truth."*

Richard Feynman — Letter to the editor of the California Tech, 1976, in the context of physics.

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In order to be more truthful in visualization we need to account for uncertainty.

Robert B. Haber and David A. McNabb defined two types of uncertainty w.r.t. visualization:
The **visualization of uncertainty** and **uncertainty of visualization**.

Uncertainty in Visualization

Uncertainty in Data

- accuracy of measurements
- biases within the data or data model
- probability of trend forecasts
- completeness of data
- confidence, variability, sampling, …

Uncertainty in Visual Display and Perception

- relevance of data w.r.t. task
- validity of measurements or data
- correctness of assumptions on data
- accuracy of visual variables
- display of accumulated data

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Uncertainty can only be mitigated to a certain extent and is highly use-case specific.

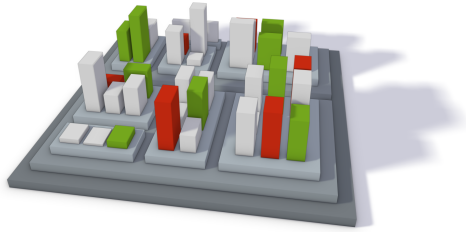
In Visual Software Analytics:

- Most metrics, especially compound metrics, such as effort, code smells etc. carry an underlying assumption which can be easily missed when reading the visualization.
- In order to facilitate knowledge-based decision making or guiding, the visualization should provide information on the relevance of any data item.

Uncertainty in Software Maps | **Attribute Uncertainty**

Example: Development activity within a software project using the number of modified lines-of-code:

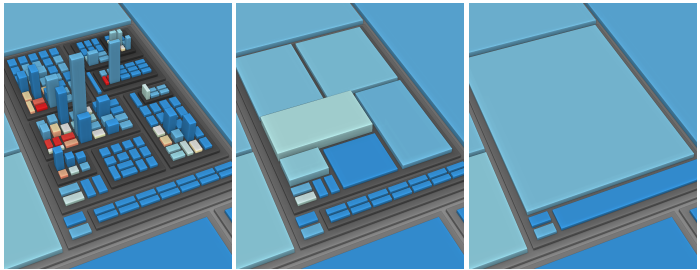
- Mapping: Number of developers is mapped to color: red = 1, gray = 2, green > 2, height is LoC.
- Task: Identification of knowledge monopolies.

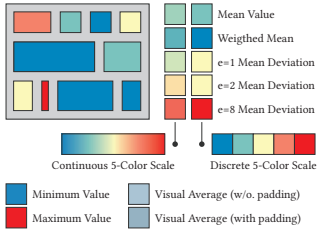


Mind the Map:

- Multiple cases where knowledge monopoly is missed, e.g., changes in non-overlapping parts, relevance of changes (e.g., fixing a typo, increment copy right year in the comments, duplicating code from other authors), etc.
- Is the mapping comparable with the same attributes mapped to other, similar software projects.
  - Comparability of attributes: normalization of metrics.
  - Comparability of visual variables: scale and intensity of height and color respectively.

Uncertainty in Software Maps | **Distribution Uncertainty**

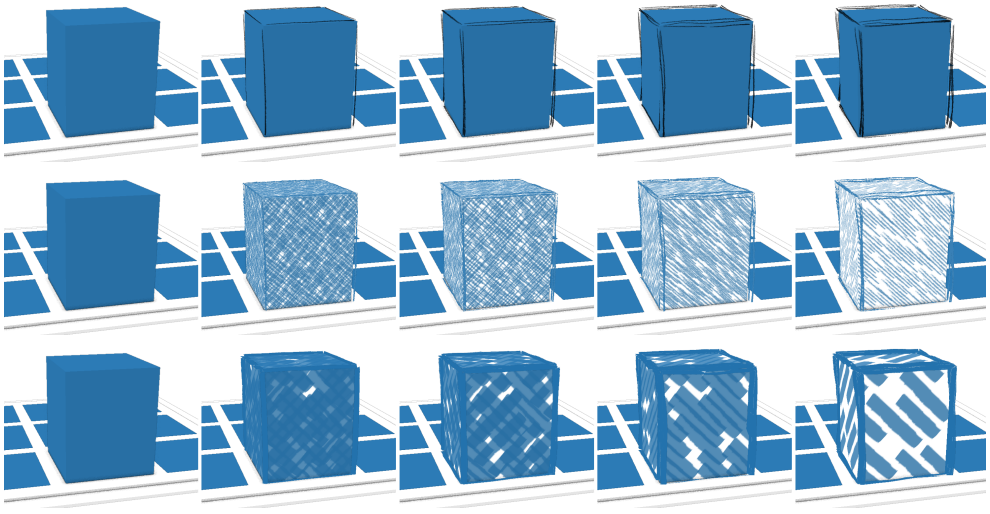




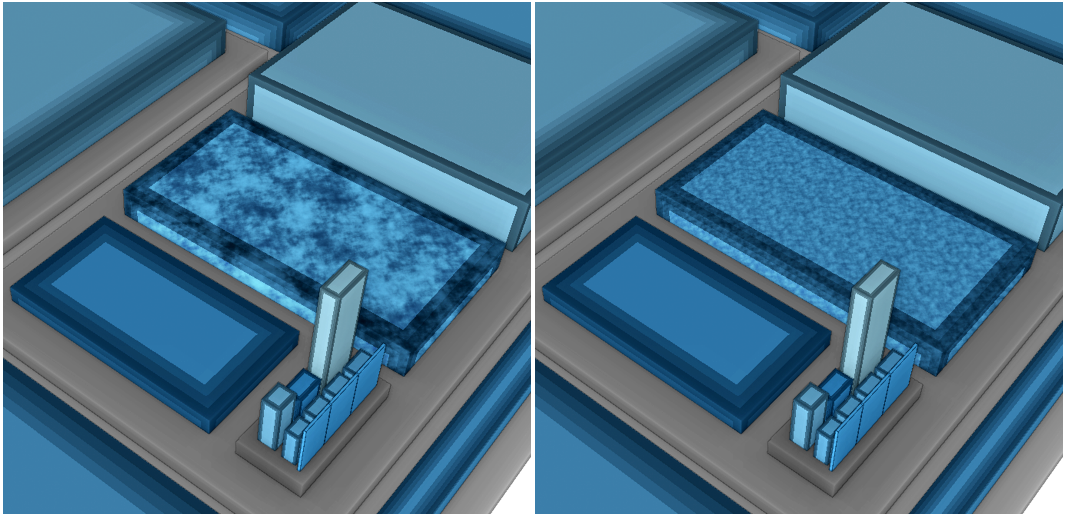
Guidelines for hierarchy aggregation (Niklas Elmqvist and Jean-Daniel Fekete, 2009) can be used for identifying and evaluating mitigation strategies:

- **G2** Visual summary: Aggregates should convey information about underlying data;
- **G3** Visual simplicity: Aggregates should be clean and simple;
- **G4** Discriminability: Aggregates should be distinguishable from data items;
- … Entity Budget, Fidelity, and Interpretability.

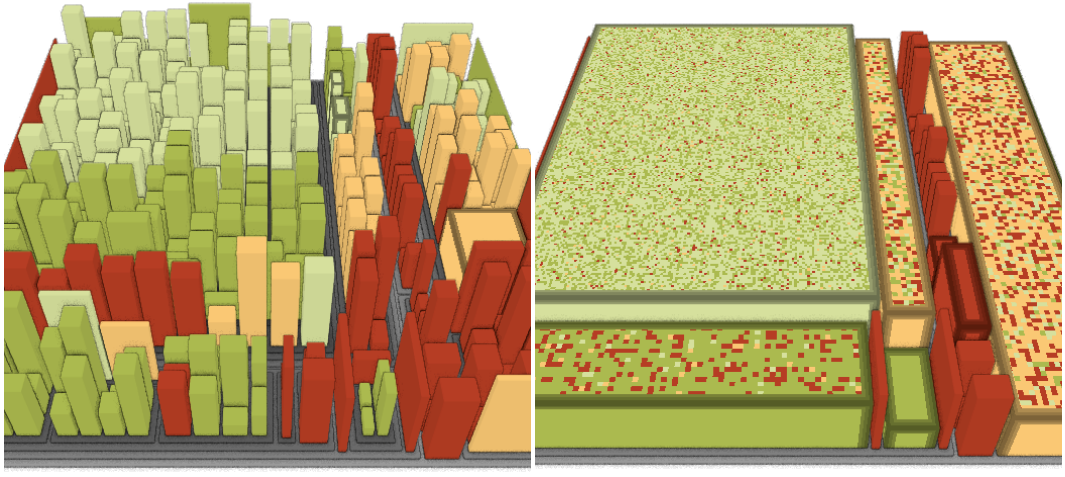
Depiction of Uncertainty | **Sketchiness**



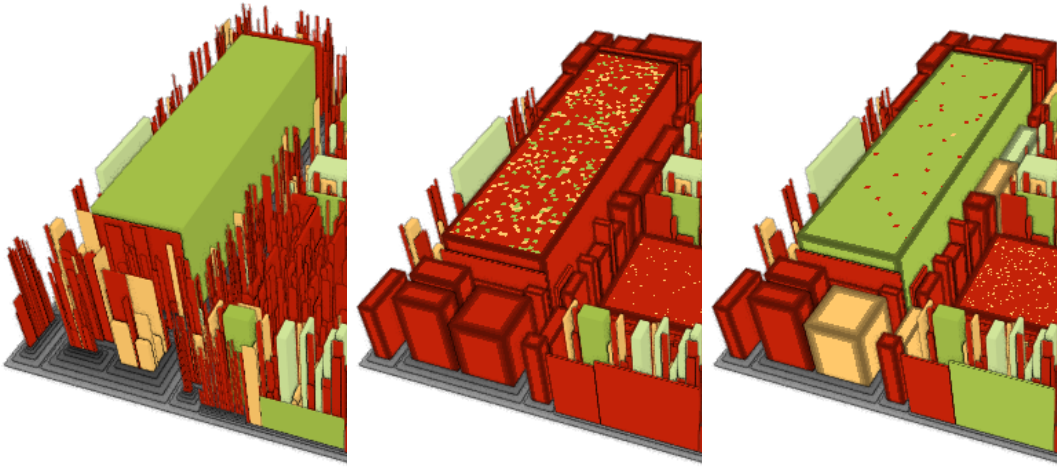
Depiction of Uncertainty | Perlin **Noise**



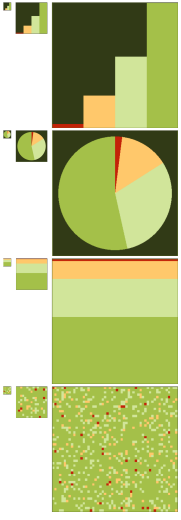
Depiction of Uncertainty | **Color Weaving**

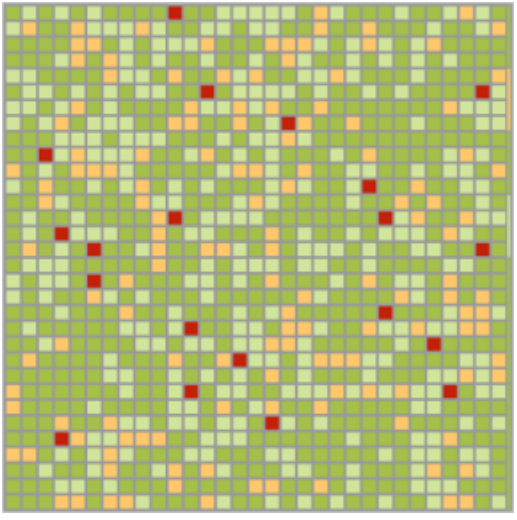


Depiction of Uncertainty | **Color Weaving** (area weighted)



Depiction of Uncertainty | **Glyph Charts**





Conclusion and Future Work

We discussed promising visual variables for superimposition in treemaps to facilitate the visual display of uncertainty in mapped attributes and aggregated/underlying value distributions.

Perceptual Questions:

- Color Weaving: what underlying distribution is expected w.r.t. area, size, and number of outliers.
- Which glyph are best suited (can be read most accurately) for small display.
- To what extent can sketchiness be superimposed to color without reducing its accuracy.

Technical Challenges:

- Glyph placement and adaptive glyph scaling.
- Adaptive size computation for minimal representative patches for color weaving.

For questions, exchange of ideas, and collaboration contact us at
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